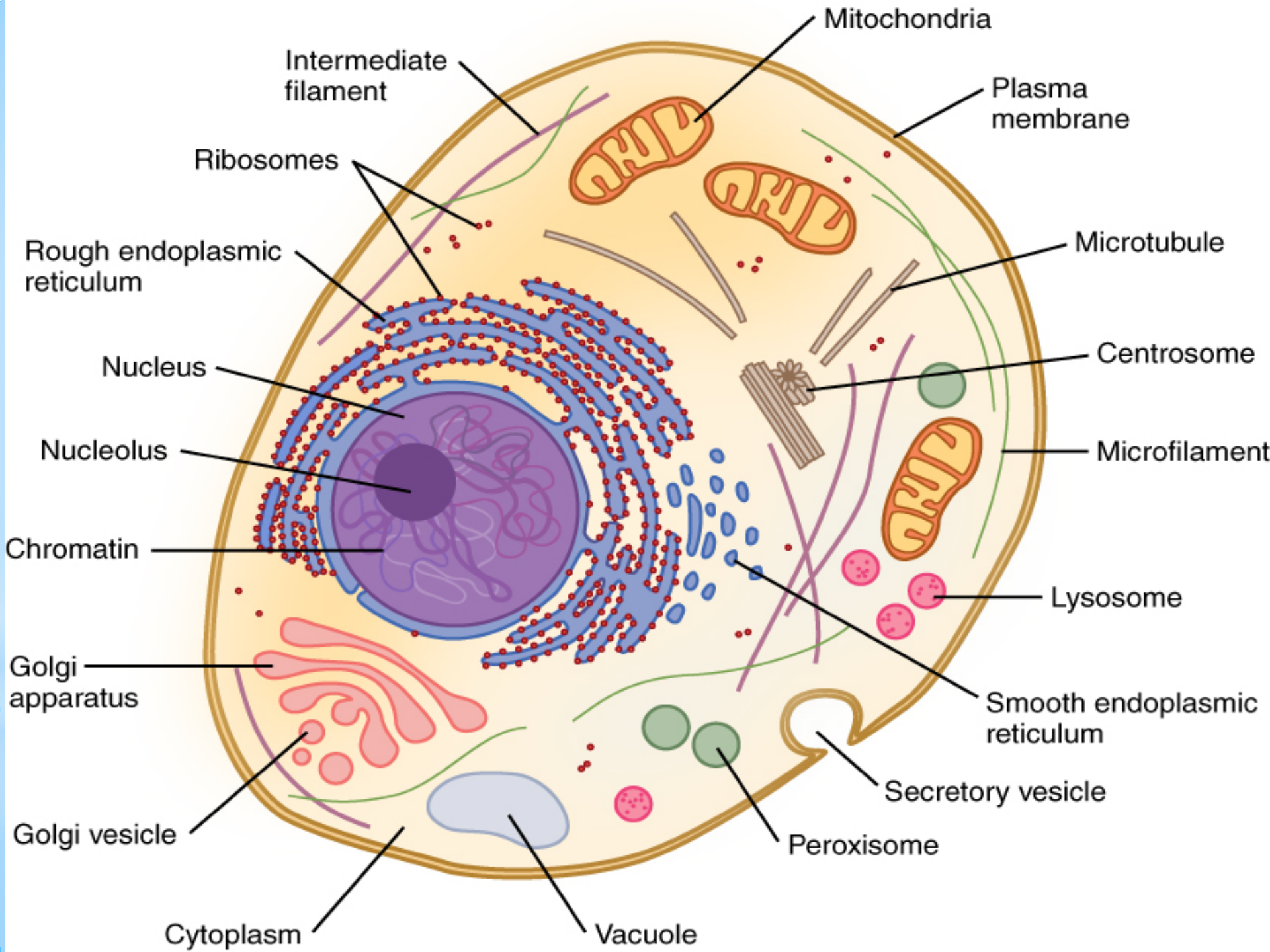
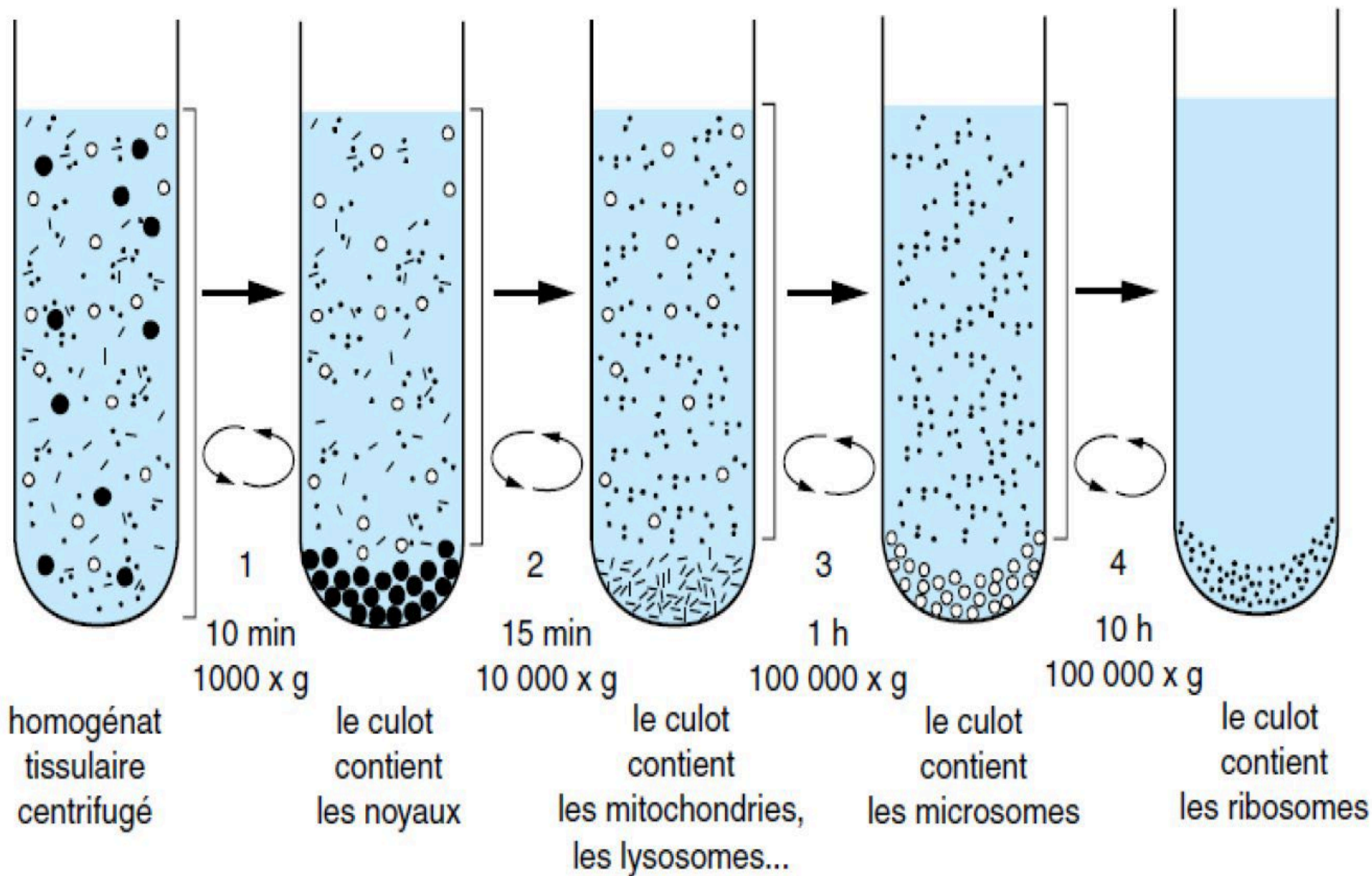


The Cytosolic Compartment





Centrifugations successives

1) basse vitesse 2) vitesse moyenne 3) vitesse élevée 4) vitesse élevée, longue durée

Relative volumes of the cytosol and organelles in a hepatocyte

Compartment	% of Cell Volume
Cytosol	54
Mitochondria	22
RER (Rough Endoplasmic Reticulum)	9
SER + Golgi (Smooth Endoplasmic Reticulum)	6
Nucleus	6
Peroxisomes	1
Lysosomes	1
Endosomes	1

Ultrastructure chemical composition

- **Water**: About 85%
- **Small molecules**: Such as: ions (Na^+ , K^+ , $\text{Ca}^{++}\dots$) and dissolved gases such as (CO_2 et O_2).
- **Medium-sized molecules**: Carbohydrates, lipids, amino acids, nucleotides, and various metabolites.
- **Macromolecules**: polysaccharides, proteins (including cytoskeleton proteins), enzymes, glycoproteins, nucleic acids...

Ionic composition of the cytosol (mM)

Ions	Cytosol	Extracellular space
K ⁺	139	4
Na ⁺	12	145
Cl ⁻	4	116
HCO ₃ ⁻	12	29
Mg ²⁺	0.8	1.5
Ca ²⁺	<0.0002	1.8

Properties of cytosol

➤ 1. pH

Varies from 7 to 7.4

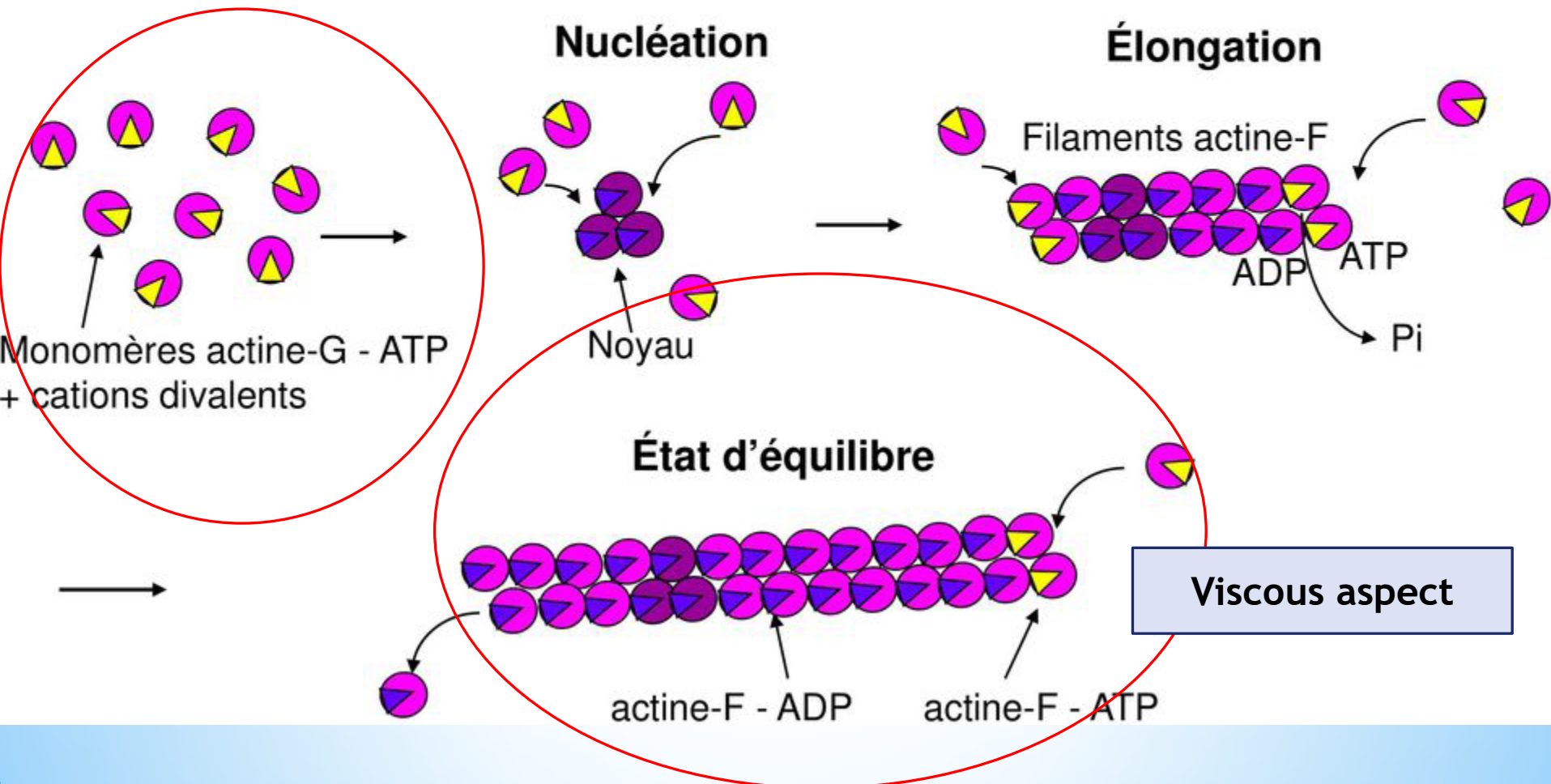
➤ 2. Viscosity

The cytosol presents two different aspects:

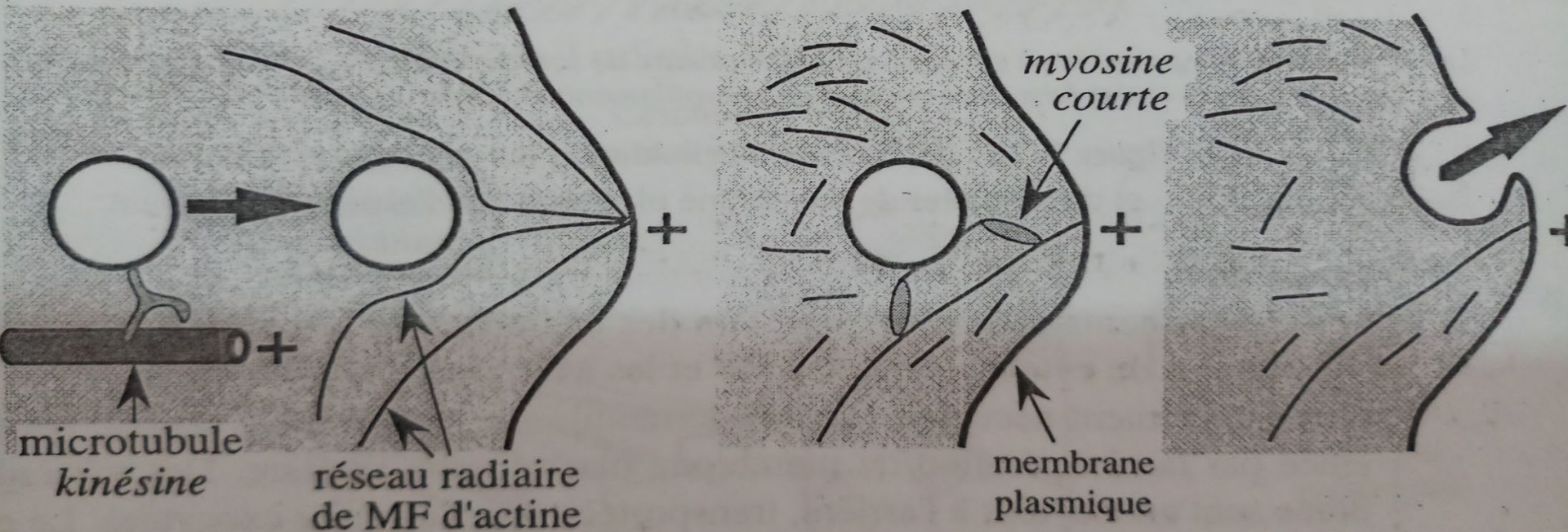
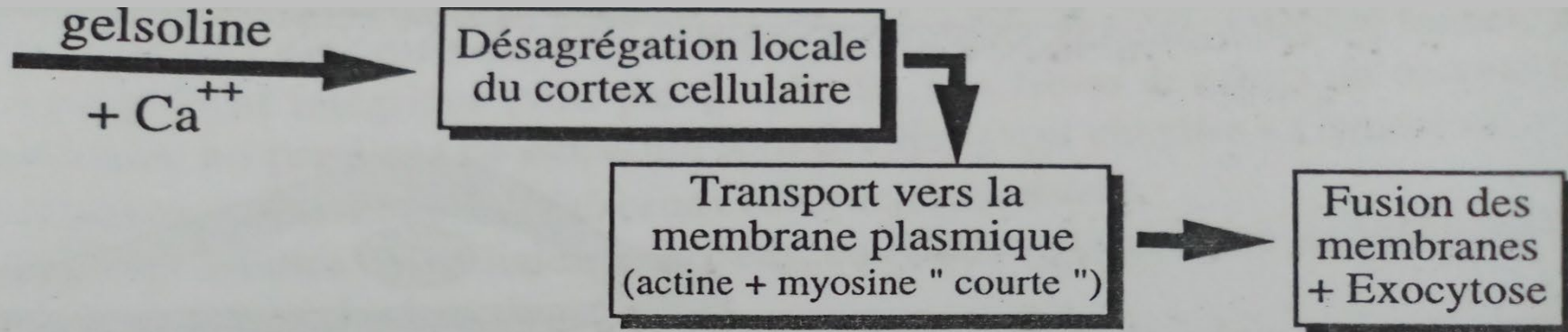
- **Viscous aspect (Gel):** when fibrous or globular proteins are linked to each other (protein polymerization).
- **Fluid aspect (Sol):** the bonds between proteins are fragmented (protein fragmentation, depolymerization).

Actin polymerization

Fluid aspect



Properties of cytosol



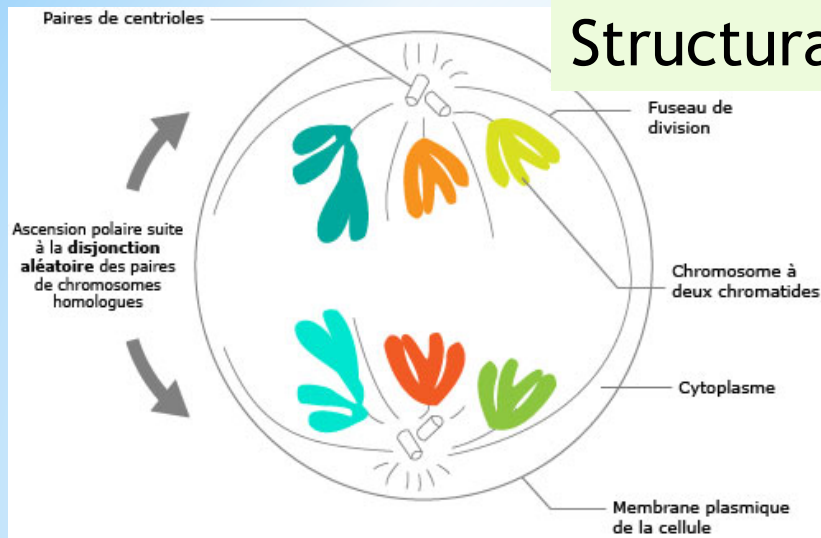
Exocytosis (fusion vesicle-membrane) requires local degradation of the cell cortex.

➤ **3. Mobility**

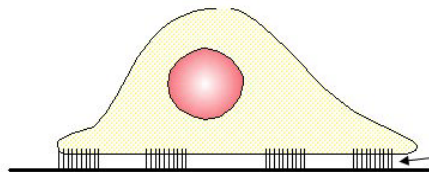
A. Circular movement :Also known as cytoplasmic streaming, this is the continuous, circular flow of the cytosol within the cell.

B. Structural movements :These movements rely on the dynamic assembly and disassembly of the cytoskeleton proteins.

Structural movements



3 grandes phases : émission d'un pseudopode, d'un filopode
adhésion du pseudopode
rétraction sur ce pseudopode

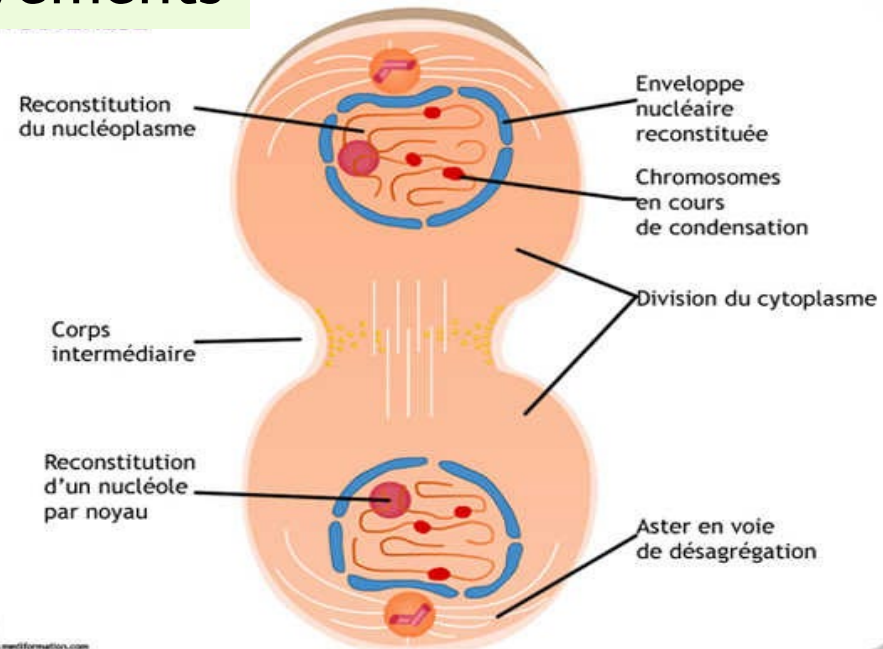
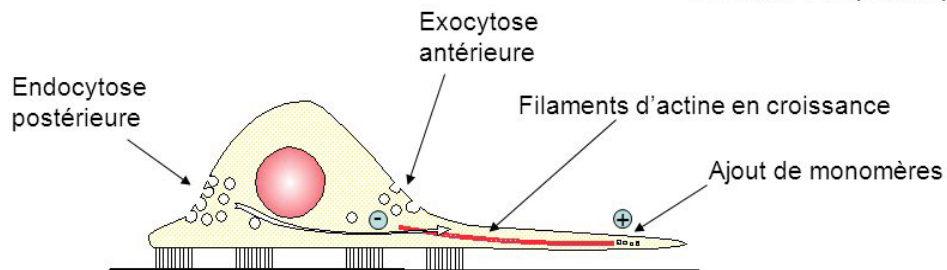


STADE INITIAL :

Cellule fixée au substrat par des plaques d'adhésion

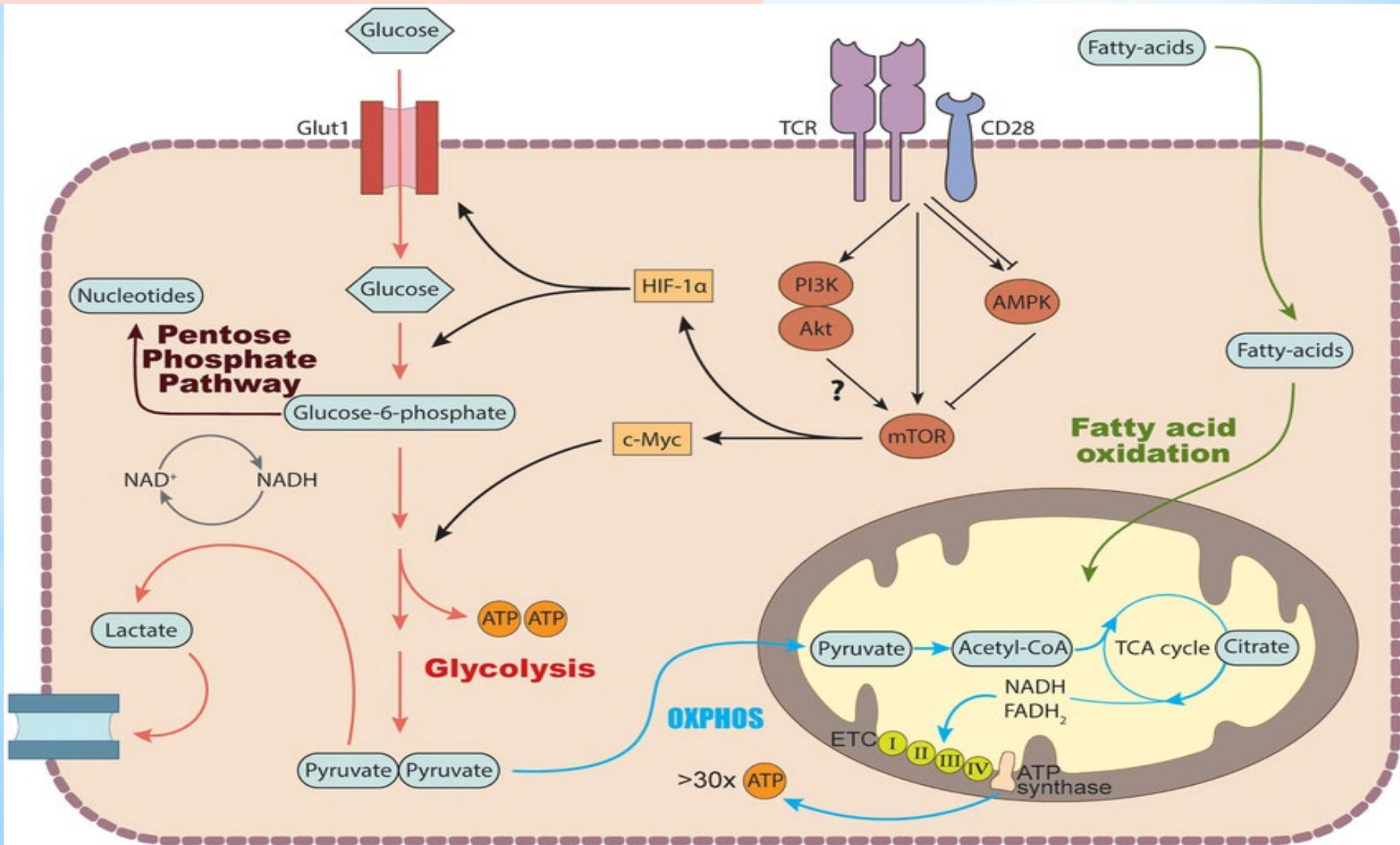
PHASE 1 :

Emission d'un pseudopode

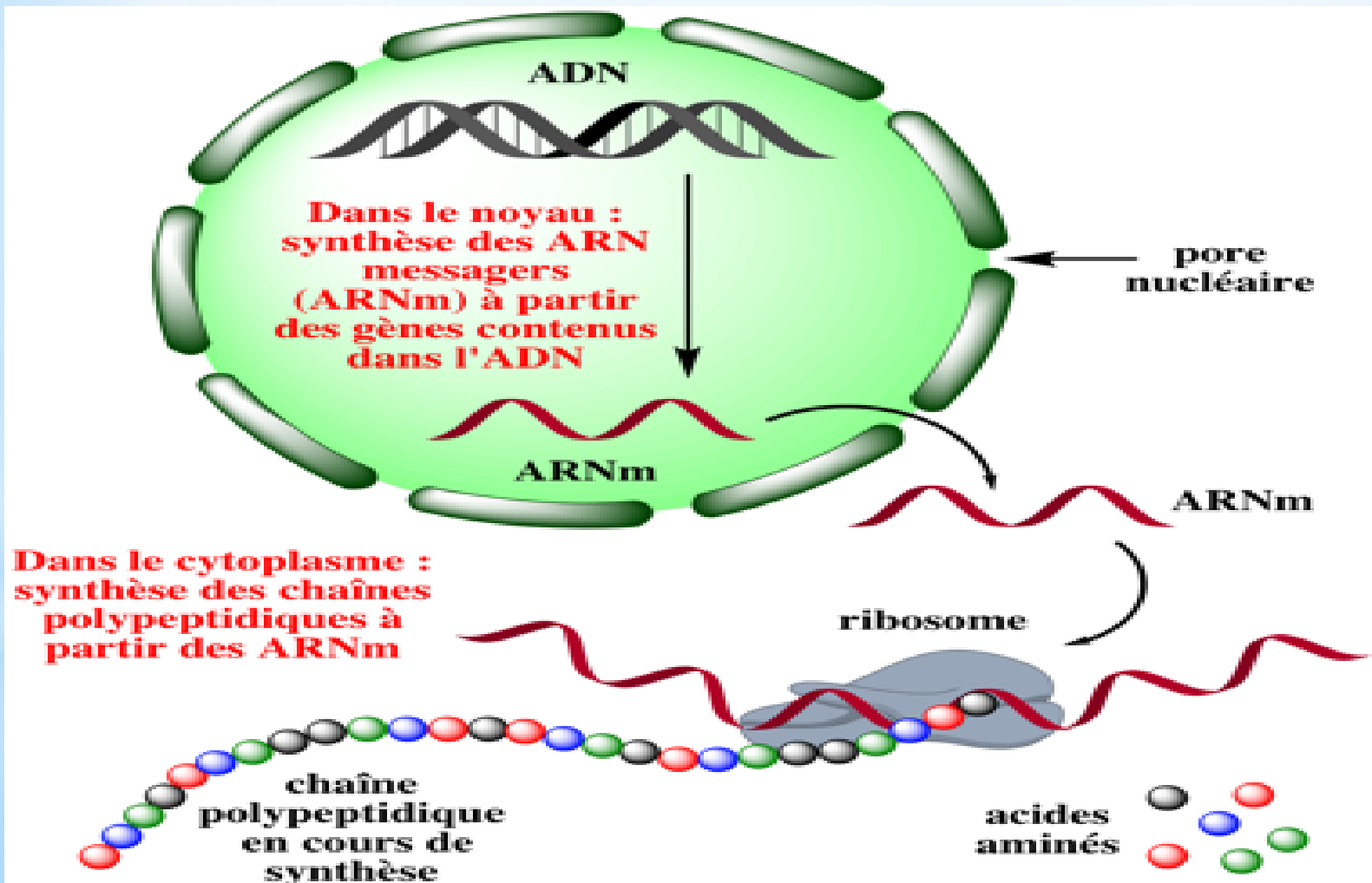


Main functions of cytosol

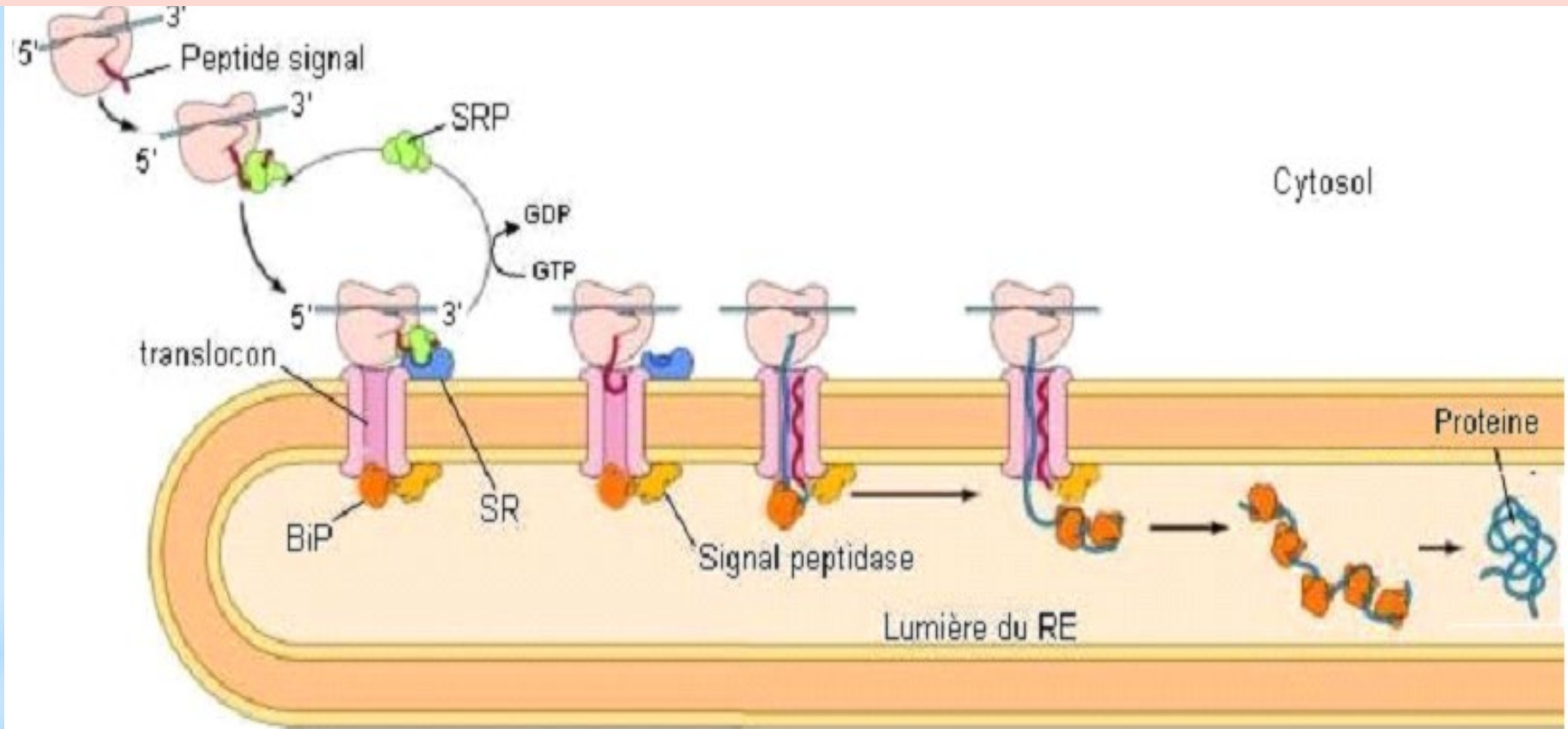
1. Metabolic Crossroads



2. Synthesis, modification, and degradation of proteins



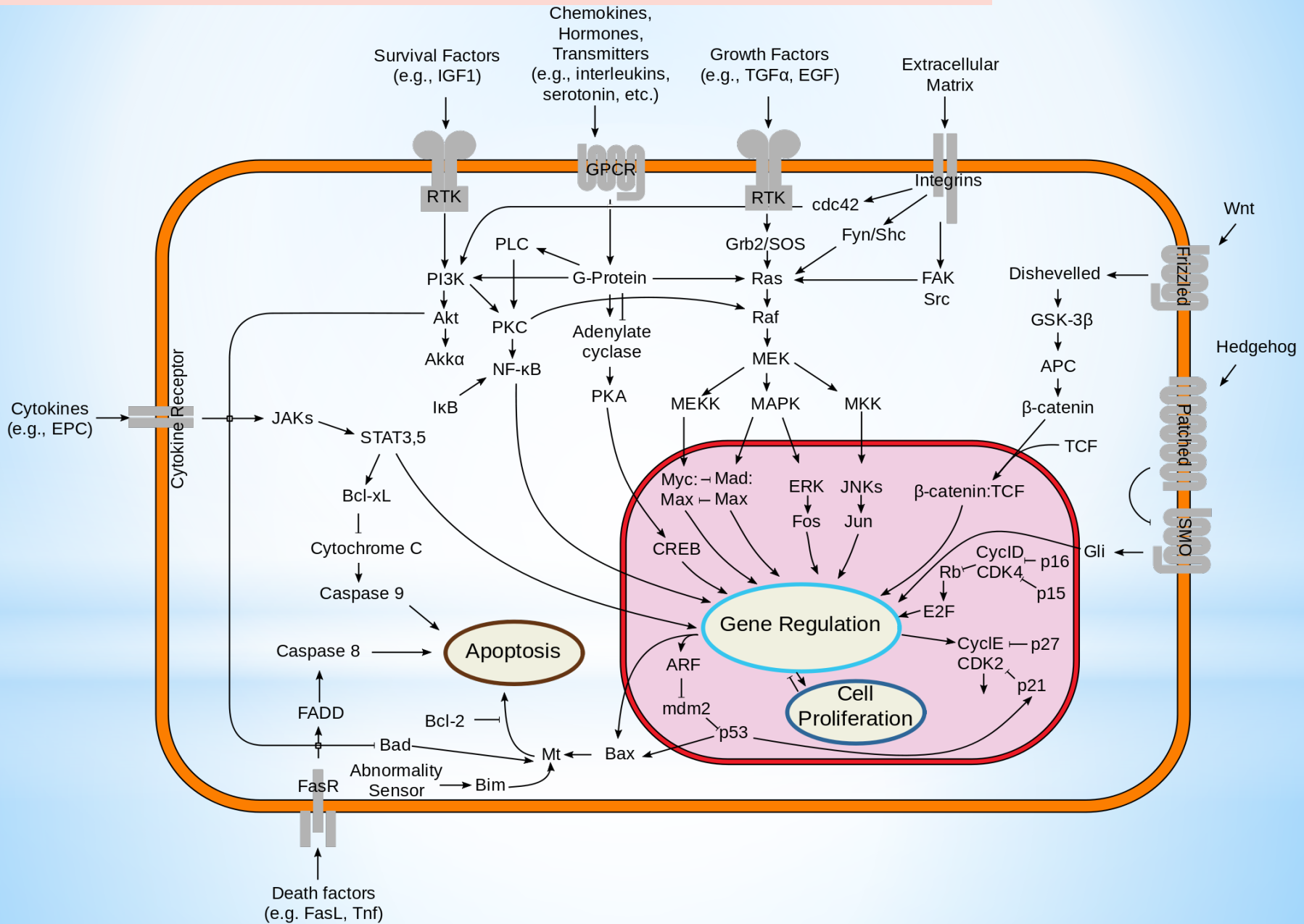
2. Synthesis, modification, and degradation of proteins



Translocation co-translationnelle des protéines
(Membranes plasmiques, Lysosomes, Vésicules sécrétoires)

Molecular Biology of the Cell

3. Signal transduction



4. Storage of energy reserves

